

LAPORAN PRAKTIKUM 13
MATA KULIAH DEVELOPMENT OPERATION
" Penerapan Testing pada DevOps dalam Pengembangan Web menggunakan
CodeIgniter Framework"



Dosen Pengampu : Hastari Utama, M.Cs

Disusun Oleh:

Nama : Muhammad Rizki Fajri

NIM : 23.01.5094

Kelas : D3-TI-03

PROGRAM STUDI D3 TEKNIK INFORMATIKA
FAKULTAS ILMU KOMPUTER
UNIVERSITAS AMIKOM YOGYAKARTA
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A. Dasar Teori

- a) Docker: Sebuah platform untuk mengembangkan, mengirim, dan menjalankan aplikasi dalam wadah (container) yang terisolasi. Docker memungkinkan kita untuk membungkus aplikasi beserta semua dependensinya, memastikan lingkungan pengembangan konsisten di antara semua pengembang.
- b) Apache2: Salah satu server web paling populer di dunia. Dalam konfigurasi ini, Apache akan bertindak sebagai entry point yang menerima permintaan dari pengguna dan meneruskannya ke layanan PHP-FPM untuk diproses.
- c) CodeIgniter 4: Sebuah framework PHP yang ringan dan cepat. CodeIgniter 4 hadir dengan testing suite yang sudah terintegrasi dengan PHPUnit, mempermudah proses penulisan tes otomatis.
- d) Test-Driven Development (TDD): Sebuah metodologi pengembangan di mana pengembang menulis tes otomatis sebelum menulis kode fungsionalnya. Alur kerjanya adalah Red -> Green -> Refactor.
- e) Unit Testing: Proses pengujian unit atau komponen terkecil dari perangkat lunak secara terisolasi (misalnya, satu metode dalam sebuah class).
- f) Integration Testing: Proses pengujian interaksi antara beberapa komponen yang terintegrasi. Dalam konteks MVC, ini berarti menguji apakah Controller, Model, dan View dapat bekerja sama dengan benar.

B. Tujuan

- a) Menyiapkan lingkungan pengembangan CodeIgniter 4 menggunakan Docker dan Docker Compose dengan server web Apache.
- b) Memahami konsep dasar Test-Driven Development (TDD), Unit Testing, dan Integration Testing.
- c) Menulis dan menjalankan Unit Test untuk komponen Model pada CodeIgniter.
- d) Menulis dan menjalankan Integration Test untuk alur MVC (Model-View-Controller).
- e) Menerapkan siklus TDD (Red-Green-Refactor) dalam pengembangan sebuah fitur

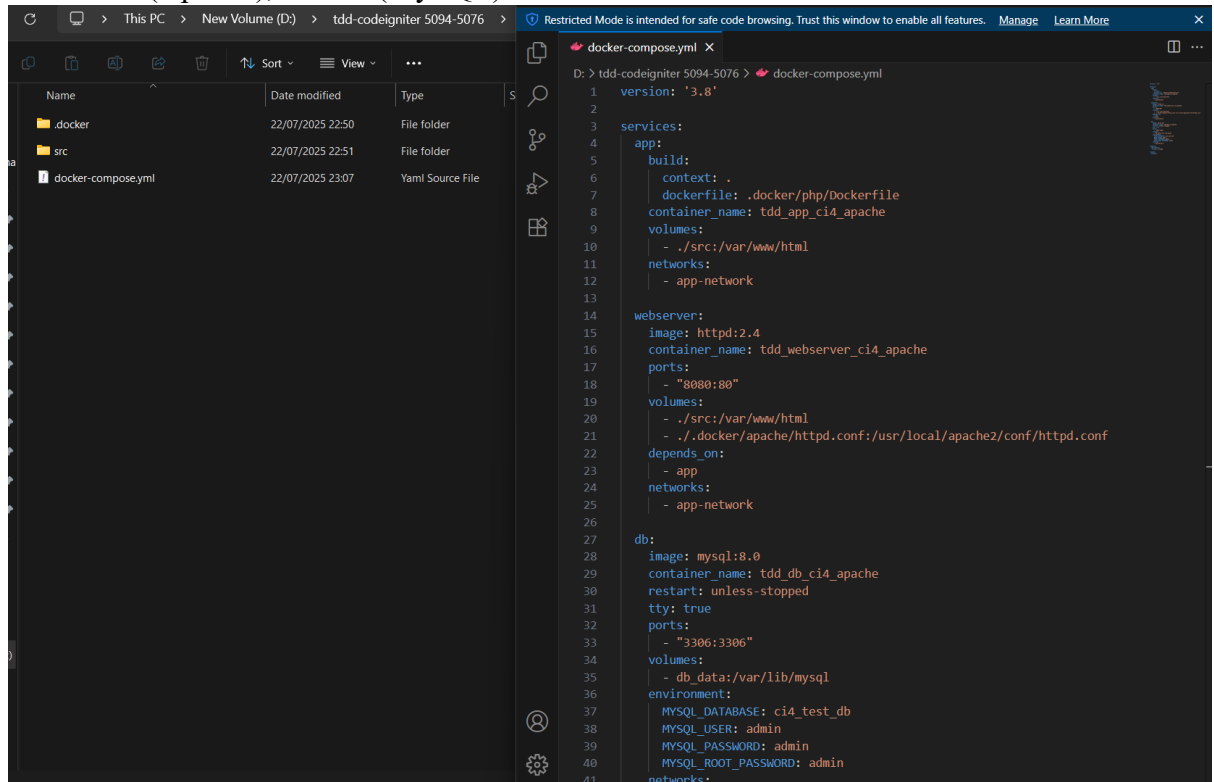
C. Langkah langkah Praktikum

- **Langkah 1: Mempersiapan Lingkungan Docker**

1. Membuat struktur direktori proyek seperti berikut

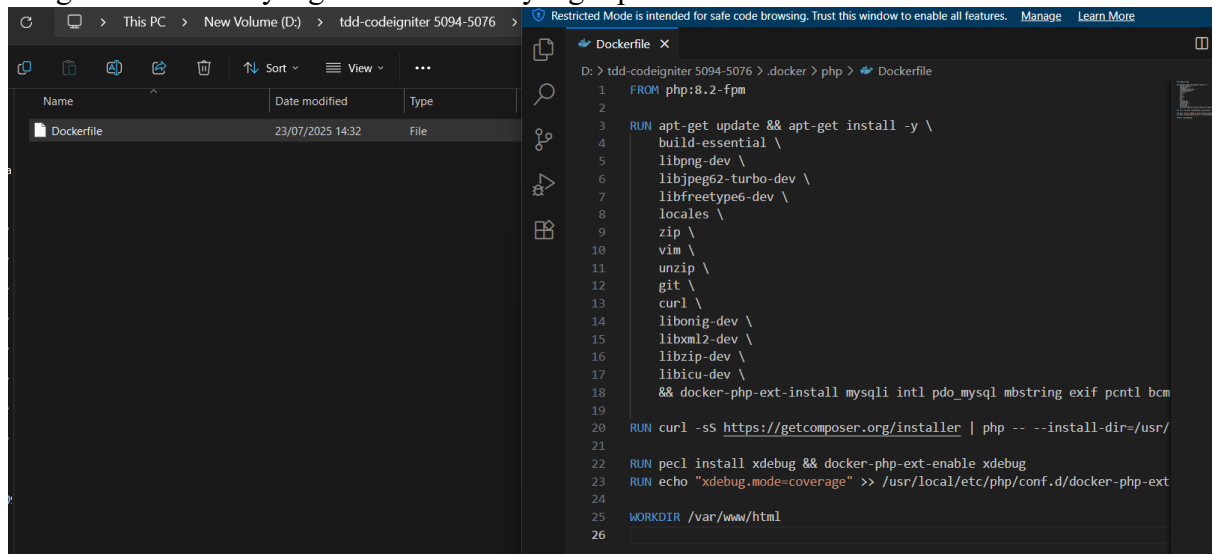
```
tdd-codeigniter/  
├── .docker/  
│   ├── apache/  
│   │   └── httpd.conf  
│   ├── php/  
│   │   └── Dockerfile  
├── src/  
└── docker-compose.yml
```

2. Isi file docker-compose.yml. File ini mendefinisikan tiga layanan: app (PHP), webserver (Apache), dan db (MySQL).



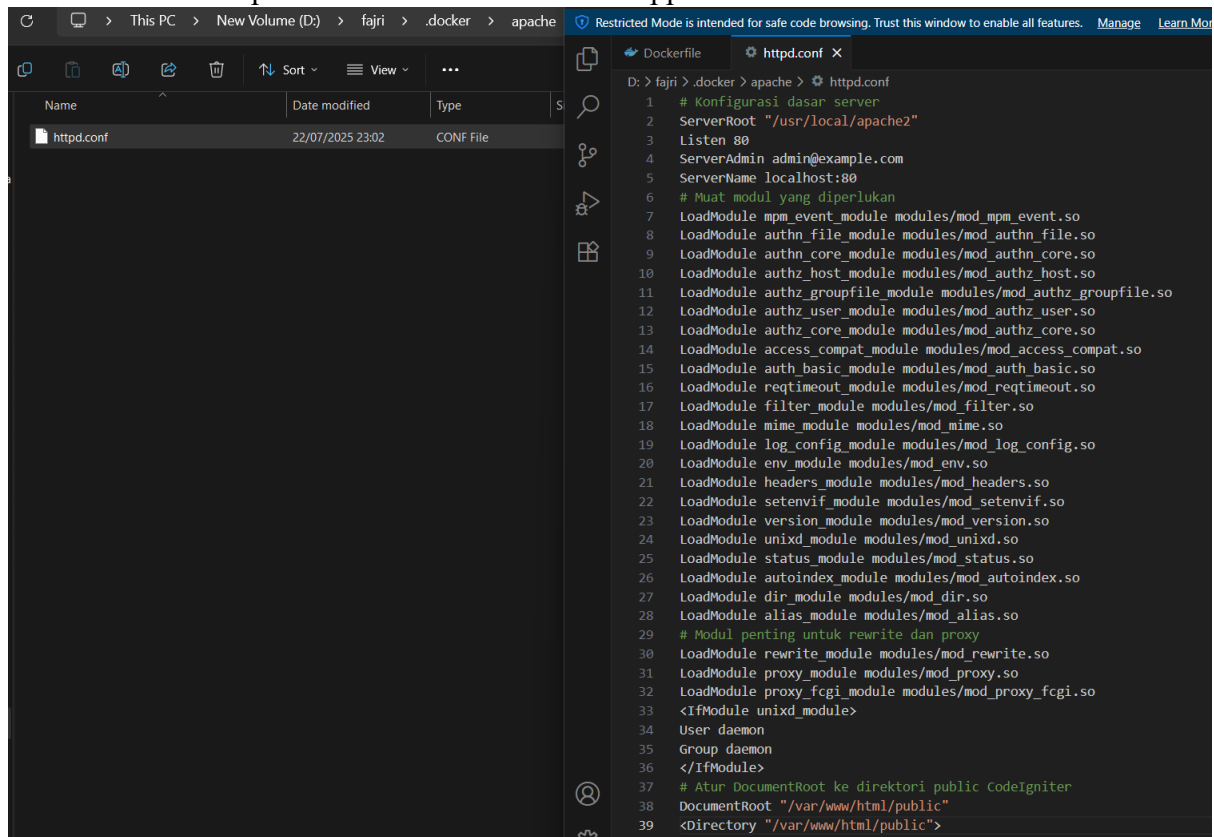
```
1 version: '3.8'  
2  
3 services:  
4   app:  
5     build:  
6       context: .  
7       dockerfile: .docker/php/Dockerfile  
8     container_name: tdd_app_ci4_apache  
9     volumes:  
10      - ./src:/var/www/html  
11     networks:  
12      - app-network  
13  
14   webserver:  
15     image: httpd:2.4  
16     container_name: tdd_webserver_ci4_apache  
17     ports:  
18      - "8080:80"  
19     volumes:  
20      - ./src:/var/www/html  
21      - ./.docker/apache/httpd.conf:/usr/local/apache2/conf/httpd.conf  
22     depends_on:  
23      - app  
24     networks:  
25      - app-network  
26  
27   db:  
28     image: mysql:8.0  
29     container_name: tdd_db_ci4_apache  
30     restart: unless-stopped  
31     tty: true  
32     ports:  
33      - "3306:3306"  
34     volumes:  
35      - db_data:/var/lib/mysql  
36     environment:  
37      MYSQL_DATABASE: ci4_test_db  
38      MYSQL_USER: admin  
39      MYSQL_PASSWORD: admin  
40      MYSQL_ROOT_PASSWORD: admin  
41     networks:
```

3. Isi file Dockerfile di `.docker/php/Dockerfile`. File ini digunakan untuk membangun image PHP kustom yang berisi ekstensi yang diperlukan.



```
D: > tdd-codeigniter 5094-5076 > .docker > php > Dockerfile
1 FROM php:8.2-fpm
2
3 RUN apt-get update && apt-get install -y \
4     build-essential \
5     libpng-dev \
6     libjpeg62-turbo-dev \
7     libfreetype6-dev \
8     locales \
9     zip \
10    vim \
11    unzip \
12    git \
13    curl \
14    libonig-dev \
15    libxml2-dev \
16    libzip-dev \
17    libicu-dev \
18    && docker-php-ext-install mysqli intl pdo_mysql mbstring exif pcntl bcmath
19
20 RUN curl -sS https://getcomposer.org/installer | php -- --install-dir=/usr/
21
22 RUN pecl install xdebug && docker-php-ext-enable xdebug
23 RUN echo "xdebug.mode=coverage" >> /usr/local/etc/php/conf.d/docker-php-ext
24
25 WORKDIR /var/www/html
26
```

4. Isi file `httpd.conf` di `.docker/apache/httpd.conf`. File ini mengonfigurasi Apache untuk meneruskan permintaan PHP ke kontainer app



```
D: > fajri > .docker > apache > httpd.conf
1 # Konfigurasi dasar server
2 ServerRoot "/usr/local/apache2"
3 Listen 80
4 ServerAdmin admin@example.com
5 ServerName localhost:80
6 # Muat modul yang diperlukan
7 LoadModule mpm_event_module modules/mod_mpm_event.so
8 LoadModule authn_file_module modules/mod_authn_file.so
9 LoadModule authn_core_module modules/mod_authn_core.so
10 LoadModule authz_host_module modules/mod_authz_host.so
11 LoadModule authz_groupfile_module modules/mod_authz_groupfile.so
12 LoadModule authz_user_module modules/mod_authz_user.so
13 LoadModule authz_core_module modules/mod_authz_core.so
14 LoadModule access_compat_module modules/mod_access_compat.so
15 LoadModule auth_basic_module modules/mod_auth_basic.so
16 LoadModule reqtimeout_module modules/mod_reqtimeout.so
17 LoadModule filter_module modules/mod_filter.so
18 LoadModule mime_module modules/mod_mime.so
19 LoadModule log_config_module modules/mod_log_config.so
20 LoadModule env_module modules/mod_env.so
21 LoadModule headers_module modules/mod_headers.so
22 LoadModule setenvif_module modules/mod_setenvif.so
23 LoadModule version_module modules/mod_version.so
24 LoadModule unixd_module modules/mod_unixd.so
25 LoadModule status_module modules/mod_status.so
26 LoadModule autoindex_module modules/mod_autoindex.so
27 LoadModule dir_module modules/mod_dir.so
28 LoadModule alias_module modules/mod_alias.so
29 # Modul penting untuk rewrite dan proxy
30 LoadModule rewrite_module modules/mod_rewrite.so
31 LoadModule proxy_module modules/mod_proxy.so
32 LoadModule proxy_fcgi_module modules/mod_proxy_fcgi.so
33 <IfModule unixd_module>
34 User daemon
35 Group daemon
36 </IfModule>
37 # Atur DocumentRoot ke direktori public CodeIgniter
38 DocumentRoot "/var/www/html/public"
39 <Directory "/var/www/html/public">
```

- ```
D:\tdd-codeigniter 5094-5076>docker-compose up --build
time="2025-07-23T17:29:16+07:00" level=warning msg="D:\\tdd-codeigniter 5094-5076\\docker-compose.yml: the attribute 'version' is deprecated, please remove it to avoid potential confusion"
Compose now can delegate build to bake for better performances
Just set COMPOSE_BAKE=true
[+] Building 69.4s (11/11) FINISHED
=> [app internal] load build definition from Dockerfile
=> => transferring dockerfile: 709B
=> [app internal] load metadata for docker.io/library/php:8.2-fpm
=> [app internal] load .dockerignore
=> => transferring context: 2B
=> [app 1/6] FROM docker.io/library/php:8.2-fpm@sha256:f07ab603daecf7010abca139bffe40d60d8569377c112c85c0594ad63
=> => resolve docker.io/library/php:8.2-fpm@sha256:f07ab603daecf7010abca139bffe40d60d8569377c112c85c0594ad632fc3
=> CACHED [app 2/6] RUN apt-get update && apt-get install -y build-essential libpng-dev libjpeg62-tu
=> CACHED [app 3/6] RUN curl -s https://getcomposer.org/installer | php -- --install-dir=/usr/local/bin --filen
=> [app 4/6] RUN pecl install xdebug && docker-php-ext-enable xdebug
=> [app 5/6] RUN echo "xdebug.mode=coverage" >> /usr/local/etc/php/conf.d/docker-php-ext-xdebug.ini
=> [app 6/6] WORKDIR /var/www/html
=> [app] exporting to image
=> => exporting layers
=> => exporting manifest sha256:89612a54814d58c45cd50cca399f1b481770d7bd8b3631a3af371202b121267e2
=> => exporting config sha256:16fc3e092a2822b5b77245101d18d4d9a9102d030cf77fedd914b86142d32f90
=> => exporting attestation manifest sha256:fd062a5a704b9428ee9a9cf639db3fa091472ba11d9dd9ca7cbbba4f8d5e1a72f
=> => exporting manifest list sha256:a70b6789f148284ff7c4388412a6e78fbbbe2fbbf0857f08c0394cd88e02406
=> => naming to docker.io/library/tdd-codeigniter5094-5076-app:latest
=> => unpacking to docker.io/library/tdd-codeigniter5094-5076-app:latest
=> [app] resolving provenance for metadata file
[+] Running 6/6
✔app Built
✔Network tdd-codeigniter5094-5076_app-network Created
✔Volume "tdd-codeigniter5094-5076_db_data" Created
✔Container tdd_app_ci4_apache Started
✔Container tdd_db_ci4_apache Started
✔Container tdd_webserver_ci4_apache Started
```

```
D:\tdd-codeigniter 5894>docker-compose exec app bash
time="2025-07-23T17:43:46+07:00" level=warning msg="D:\\tdd-codeigniter 5894-5876\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
root@555cd3d4862d:/var/www/html# rootfs-container_id=/var/www/html#
rootfs-container_id: No such file or directory
root@555cd3d4862d:/var/www/html# composer create-project codeigniter4/apstarter .
bash: composer: command not found
root@555cd3d4862d:/var/www/html# curl -sS https://getcomposer.org/installer | php -- --install-dir=/usr/local/bin --filename=composer
All settings correct for using Composer
Downloading ..

Composer (version 2.8.10) successfully installed to: /usr/local/bin/composer
Use it: php /usr/local/bin/composer

root@555cd3d4862d:/var/www/html# composer create-project codeigniter4/apstarter .
Creating a "codeigniter4/apstarter" project at ..
Installing codeigniter4/apstarter (v4.6.1)
 - Downloading codeigniter4/apstarter (v4.6.1)
 - Installing codeigniter4/apstarter (v4.6.1): Extracting archive
Created project in /var/www/html/
Loading composer repositories with package information
Updating dependencies
Lock file operations: 23 installs, 0 updates, 0 removals
 - Locking codeigniter4/framework (v4.6.1)
 - Locking fakerphp/faker (v1.24.1)
 - Locking laminas/laminas-escaper (2.17.0)
 - Locking wilkey179/vfstream (v1.6.122)
 - Locking myclabs/deep-copy (1.13.3)
 - Locking nikic/php-parser (v5.5.0)
 - Locking phar-io/manifest (2.0.4)
 - Locking phar-io/version (3.2.1)
 - Locking phpunit/php-code-coverage (10.1.16)
 - Locking phpunit/php-file-iterator (4.0.0)
 - Locking phpunit/php-invoker (4.0.0)
 - Locking phpunit/php-text-template (3.0.1)
 - Locking phpunit/php-timer (6.0.0)
 - Locking phpunit/phpunit (10.5.48)
 - Locking psr/container (2.0.2)
```

```
D:\tdd-codeigniter 5094-5076>cd src
D:\tdd-codeigniter 5094-5076\src>copy env .env
1 file(s) copied.
D:\tdd-codeigniter 5094-5076\src>cd ..
```

|               |                  |                       |       |
|---------------|------------------|-----------------------|-------|
| .env          | 02/05/2025 23:00 | ENV File              | 3 KB  |
| .gitignore    | 02/05/2025 23:00 | Git Ignore Source ... | 2 KB  |
| builds        | 02/05/2025 23:00 | File                  | 4 KB  |
| composer.json | 02/05/2025 23:00 | JSON Source File      | 2 KB  |
| composer.lock | 23/07/2025 17:48 | LOCK File             | 75 KB |
| env           | 02/05/2025 23:00 | File                  | 3 KB  |

3. Buka file src\.env dengan editor teks dan sesuaikan konfigurasinya seperti pada prosedur sebelumnya

```

8
9 CI_ENVIRONMENT = testing
10
11 #-----
12 # APP
13 #-----
14
15 # app.baseUrl = ''
16 # If you have trouble with `.` , you could also use ` _ ` .
17 # app.baseUrl = ''
18 # app.forceGlobalSecureRequests = false
19 # app.CSPEnabled = false
20
21 #-----
22 # DATABASE
23 #-----
24
25 database.default.hostname = db
26 database.default.database = ci4_test_db
27 database.default.username = admin
28 database.default.password = admin
29 database.default.DBDriver = MySQLi
30 database.default.DBPrefix =
31 database.default.port = 3306
32
33 # If you use MySQLi as tests, first update the values of
34
35 database.tests.hostname = db
36 database.tests.database = ci4_test_db
37 database.tests.username = admin
38 database.tests.password = admin
39 database.tests.DBDriver = MySQLi
40 database.tests.DBPrefix =
41 database.tests.charset = utf8mb4
42 database.tests.Collat = utf8mb4_general_ci
43 database.tests.port = 3306
44

```

4. Masuk ke dalam terminal kontainer app dari PowerShell/CMD: docker-compose exec app bash

```

D:\tdd-codeigniter 5094-5076>docker-compose exec app bash
time="2025-07-25T10:13:09+07:00" level=warning msg="D:\\tdd-codeigniter 5094-5076\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"

```

5. Dari dalam kontainer, buat dan jalankan migrasi database:

```

root@d0877f22f84c:/var/www/html# php spark make:migration create_products_table

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 03:13:47 UTC+00:00
File created: APPPATH/Database/Migrations/2025-07-25-031354_CreateProductsTable.php

root@d0877f22f84c:/var/www/html#

```

6. Mengisi File Migrasi

```

root@d0877f22f84c:/var/www/html/app/Database/Migrations# vi 2025-07-25-031354_CreateProductsTable.php
root@d0877f22f84c:/var/www/html/app/Database/Migrations#

```

```

C:\Windows\System32\cmd.e x + v
<?php
namespace App\Database\Migrations;
use CodeIgniter\Database\Migration;

class CreateProductsTable extends Migration
{
 public function up()
 {
 // Mendefinisikan kolom-kolom untuk tabel 'products'
 $this->forge->addField([
 'id' => [
 'type' => 'INT',
 'constraint' => 5,
 'unsigned' => true,
 'auto_increment' => true,
],
 'name' => [
 'type' => 'VARCHAR',
 'constraint' => '100',
],
 'price' => [
 'type' => 'DECIMAL',
 'constraint' => '10,2', // Untuk menyimpan nilai uang
],
 // Anda bisa menambahkan kolom created_at dan updated_at jika perlu
 // 'created_at' => [
 // 'type' => 'DATETIME',
 // 'null' => true,
 //],
 // 'updated_at' => [
 // 'type' => 'DATETIME',
 // 'null' => true,
 //],
]);
 }
};

```

## 7. Mengatur nama tabel yang dituju tanpa prefix.

|                           |                  |                 |   |     |     |                                                        |
|---------------------------|------------------|-----------------|---|-----|-----|--------------------------------------------------------|
| Constants.php             | 02/05/2025 23:00 | PHP Source File | 4 | 164 | 165 | public array \$tests = [                               |
| ContentSecurityPolicy.php | 02/05/2025 23:00 | PHP Source File | 5 | 166 | 167 | 'dsn' => '',                                           |
| Cookie.php                | 02/05/2025 23:00 | PHP Source File | 4 | 168 | 169 | 'hostname' => '127.0.0.1',                             |
| Cors.php                  | 02/05/2025 23:00 | PHP Source File | 4 | 170 | 171 | 'username' => '',                                      |
| CURLRequest.php           | 02/05/2025 23:00 | PHP Source File | 1 | 172 | 173 | 'password' => '',                                      |
| Database.php              | 25/07/2025 10:37 | PHP Source File | 7 | 174 | 175 | 'database' => ':memory:',                              |
| DocTypes.php              | 02/05/2025 23:00 | PHP Source File | 3 | 176 | 177 | 'DBDriver' => 'SQLite3',                               |
| Email.php                 | 02/05/2025 23:00 | PHP Source File | 3 | 178 | 179 | 'DBPrefix' => '', // Needed to ensure we're working co |
| Encryption.php            | 02/05/2025 23:00 | PHP Source File | 3 | 180 | 181 | 'pconnect' => false,                                   |
| Events.php                | 02/05/2025 23:00 | PHP Source File | 2 | 182 | 183 | 'DBDebug' => true,                                     |
| Exceptions.php            | 02/05/2025 23:00 | PHP Source File | 4 | 184 | 185 | 'charset' => 'utf8',                                   |
| Feature.php               | 02/05/2025 23:00 | PHP Source File | 1 | 186 | 187 | 'dbcollat' => '',                                      |
| Filters.php               | 02/05/2025 23:00 | PHP Source File | 4 | 188 | 189 | 'swapPre' => '',                                       |
| ForeignCharacters.php     | 02/05/2025 23:00 | PHP Source File | 1 | 190 | 191 | 'encrypt' => false,                                    |
| Format.php                | 02/05/2025 23:00 | PHP Source File | 3 |     |     | 'compress' => false,                                   |
| Generators.php            | 02/05/2025 23:00 | PHP Source File | 3 |     |     | 'stricton' => false,                                   |
| Honeypot.php              | 02/05/2025 23:00 | PHP Source File | 1 |     |     | 'failover' => [],                                      |
|                           |                  |                 |   |     |     | 'port' => 3306,                                        |
|                           |                  |                 |   |     |     | 'foreignKeys' => true,                                 |
|                           |                  |                 |   |     |     | 'busyTimeout' => 1000,                                 |
|                           |                  |                 |   |     |     | 'dateFormat' => [                                      |
|                           |                  |                 |   |     |     | 'date' => 'Y-m-d',                                     |
|                           |                  |                 |   |     |     | 'datetime' => 'Y-m-d H:i:s',                           |
|                           |                  |                 |   |     |     | 'time' => 'H:i:s',                                     |
|                           |                  |                 |   |     |     | ],                                                     |
|                           |                  |                 |   |     |     | ];                                                     |

## 8. Menjalankan Migrasi

```

root@d0877f22f84c:/var/www/html# php spark migrate

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 03:42:52 UTC+00:00

Running all new migrations...
Running: (App) 2025-07-25-031354_App\Database\Migrations\CreateProductsTable
Migrations complete.
root@d0877f22f84c:/var/www/html#

```

## D. Studi Kasus

### • Unit Testing untuk ProductModel (Panduan Detail)

#### Tahap 1: RED - Menulis Tes yang Gagal

1. Membuat File Tes Dari dalam terminal container

```
root@d0877f22f84c:/var/www/html# php spark make:test Models/ProductModelTest

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 03:46:15 UTC+00:00

File created: ROOTPATH/tests/Models/ProductModelTest.php
```

2. Tulis Kode Tes Buka file ProductModelTest.php tersebut dan ganti seluruh isinya dengan kode seperti berikut. Kode ini mendefinisikan kasus uji untuk mengambil semua produk.

```
root@d0877f22f84c:/var/www/html# vi ProductTest.php
```

```
C:\Windows\System32\cmd.e X + v

<?php

namespace Tests\Unit;

use CodeIgniter\Test\CIUnitTestCase;
use CodeIgniter\Test\DatabaseTestTrait;
use App\Models\ProductModel;

final class ProductTest extends CIUnitTestCase
{
 use DatabaseTestTrait;

 protected $seed = false;
 protected $basePath = APPPATH . 'Database/';
 protected $namespace = 'App';

 public function testInsertAndRetrieveProduct()
 {
 $product = new ProductModel();

 $data = [
 'name' => 'Produk Uji Coba',
 'price' => 15000.00,
];

 $product->insert($data);

 $result = $product->where('name', 'Produk Uji Coba')->first();

 $this->assertIsArray($result);
 $this->assertEquals('Produk Uji Coba', $result['name']);
 $this->assertEquals(15000.00, (float)$result['price']);
 }
}
```

3. Jalankan Tes dan Verifikasi Kegagalan Kembali ke terminal kontainer dan jalankan PHPUnit hanya untuk file tes yang baru kita buat.

```
root@d0877f22f84c:/var/www/html# php vendor/bin/phpunit tests/Unit/ProductTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Test file "tests/Unit/ProductTest.php" not found
```



## Tahap 2: GREEN - Menulis Kode Minimal untuk Lulus Tes

1. Buat File Model Gunakan spark untuk membuat file ProductModel.

```
root@d0877f22f84c:/var/www/html# php spark make:model ProductModel

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 04:07:53 UTC+00:00
File created: APPPATH/Models/ProductModel.php
```

2. Konfigurasi Model Buka file ProductModel.php dan konfigurasikan seperti berikut. Ini adalah konfigurasi minimal yang dibutuhkan oleh CodeIgniter untuk menghubungkan model ke tabel database.

```
root@d0877f22f84c:/var/www/html# vi ProductModel.php

C:\Windows\System32\cmd.e X + v

<?php

namespace App\Models;

use CodeIgniter\Model;

class ProductModel extends Model
{
 // Nama tabel di database
 protected $table = 'products';

 // Primary key dari tabel
 protected $primaryKey = 'id';

 // Kolom yang diizinkan untuk diisi (insert/update)
 protected $allowedFields = ['name', 'price'];
}
```

3. Jalankan Tes Kembali Sekarang, jalankan lagi perintah tes yang sama.

```
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Models/ProductModelTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

.
1 / 1 (100%)

Time: 00:07.118, Memory: 16.00 MB

OK (1 test, 3 assertions)

Generating code coverage report in PHP format ... done [00:00.028]
Generating code coverage report in Clover XML format ... done [00:01.357]
Generating code coverage report in HTML format ... done [00:10.823]

Code Coverage Report:
 2025-07-25 06:26:11

Summary:
 Classes: 16.67% (1/6)
 Methods: 12.50% (1/8)
 Lines: 5.26% (5/95)

Config\Database
 Methods: 100.00% (1/ 1) Lines: 100.00% (3/ 3)
```

Keberhasilan ini menandakan bahwa kode yang baru kita tulis telah memenuhi semua assertion (penegasan) yang kita definisikan di dalam tes. Fungsionalitas dasar kita sudah bekerja. Tahap GREEN berhasil.

### Tahap 3: REFACTOR - Membersihkan dan Memperbaiki Kode

1. Jalankan Tes untuk Memastikan Tidak Ada yang Rusak Meskipun kita tidak melakukan perubahan kode, praktik yang baik setelah tahap refactor adalah menjalankan seluruh rangkaian tes untuk memastikan perubahan kita tidak merusak bagian lain dari aplikasi.

```
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Controllers/ProductControllerTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Test file "tests/app/Controllers/ProductControllerTest.php" not found
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

RR.....

Time: 00:09.808, Memory: 16.00 MB

There were 2 risky tests:

1) Tests\Controllers\ProductControllerTest::testExample
This test did not perform any assertions

/var/www/html/tests/Controllers/ProductControllerTest.php:14

2) Tests\Models\ProductModelTest::testExample
This test did not perform any assertions

/var/www/html/tests/Models/ProductModelTest.php:14

OK, but there were issues!
Tests: 8, Assertions: 10, Risky: 2.

Generating code coverage report in PHP format ... done [00:00.035]
Generating code coverage report in Clover XML format ... done [00:01.316]
Generating code coverage report in HTML format ... done [00:13.301]

Code Coverage Report:
 2025-07-25 06:46:41

Summary:
Classes: 16.67% (1/6)
Methods: 25.00% (2/8)
Lines: 6.32% (6/95)

App\Database\Migrations\CreateProductsTable
App\Database\Migrations\CreateProductsTable
Methods: 50.00% (1/ 2) Lines: 3.57% (1/ 28)
Config\Database
Methods: 100.00% (1/ 1) Lines: 100.00% (3/ 3)
```

- **Integration Testing untuk ProductController (Panduan Detail)**  
**Tahap 1: RED - Menulis Tes yang Gagal**

1. Buat File Tes Dari dalam terminal kontainer, jalankan perintah spark untuk membuat kerangka file tes Controller

```
root@d0877f22f84c:/var/www/html# php spark make:test Controllers/ProductControllerTest

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 06:35:10 UTC+00:00
File created: ROOTPATH/tests/Controllers/ProductControllerTest.php
```

2. Tulis Kode Tes Integrasi Buka file ProductControllerTest.php dan ganti seluruh isinya dengan kode berikut

```
root@d0877f22f84c:/var/www/html# vi ProductControllerTest.php

<?php

namespace App\Controllers;

use CodeIgniter\Test\CIUnitTestCase;
use CodeIgniter\Test\ControllerTestTrait; // Trait untuk simulasi HTTP request
use CodeIgniter\Test\DatabaseTestTrait;
use App\Models\ProductModel;
use App\Controllers\Product;

class ProductControllerTest extends CIUnitTestCase
{
 use ControllerTestTrait, DatabaseTestTrait;

 protected $migrate = false;

 public function testIndexShowsProducts()
 {
 // ARRANGE: Siapkan data dummy di database
 $model = new ProductModel();
 $model->insert([
 'name' => 'Kamera C',
 'price' => 750.00
]);

 // ACT: Simulasikan request GET ke URL /products
 $result = $this->withURI('http://localhost:8080/products')
 ->controller(Product::class)
 ->execute('index');

 // ASSERT:
 // 1. Cek status response 200 OK
 $this->assertTrue($result->isOk());

 // 2. Pastikan terdapat teks "Daftar Produk"
 $this->assertStringContainsString('Daftar Produk', $result->getBody());

 // 3. Pastikan nama produk "Kamera C" tampil
 $this->assertStringContainsString('Kamera C', $result->getBody());
 }
}
```

3. Jalankan Tes dan Verifikasi Kegagalan Kembali ke terminal kontainer dan jalankan PHPUnit untuk file tes ini

```
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Controllers/ProductControllerTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Test file "tests/app/Controllers/ProductControllerTest.php" not found
```

Ini adalah kegagalan yang kita harapkan. Tes kita gagal karena ProductController belum kita buat. Tahap RED berhasil.

## Tahap 2: GREEN - Menulis Kode Minimal untuk Lulus Tes

1. **Membuat Controller** Gunakan spark untuk membuat file Controller. Opsi `--restful` akan membuat kerangka metode standar (index, show, create, dll.).

```
root@d0877f22f84c:/var/www/html# php spark make:controller Product --restful

CodeIgniter v4.6.1 Command Line Tool - Server Time: 2025-07-25 06:52:41 UTC+00:00
File created: APPPATH/Controllers/Product.php
```

Buka file `app/Controllers/Product.php` dan isi metode `index()`

```
root@d0877f22f84c:/var/www/html# vi app/Controllers/Product.php
```

```
<?php

namespace App\Controllers;

use App\Models\ProductModel;
use CodeIgniter\RESTful\ResourceController;

class Product extends ResourceController
{
 public function index()
 {
 $model = new ProductModel();
 $data = [
 'title' => 'Daftar Produk',
 'products' => $model->findAll()
];

 return view('products/index', $data);
 }
}
```

2. **Buat View** Buat folder products di dalam **src/app/Views/**. Kemudian, buat file baru di dalamnya bernama **index.php**. Lokasi: **src/app/Views/products/index.php** Isi dengan kode HTML sederhana

The image shows a Windows File Explorer window with the path `D:\> tdd-codeigniter 5094-5076 > src > app > Views > products`. It displays a file named `index.php` with a size of 1 KB, modified on 25/07/2025 at 13:59. Below the file explorer, a VS Code editor window is open, showing the content of `index.php`. The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <title><?= esc($title) ?></title>
6 </head>
7 <body>
8 <h1><?= esc($title) ?></h1>
9
10 <?php foreach ($products as $product): ?>
11 <?= esc($product['name']) ?> - $<?= esc($product['p
12 ?>
13 <?php endforeach; ?>
14
15 </body>
16 </html>
```

3. **Tambahkan Route** Buka file **src/app/Config/Routes.php** dan tambahkan baris berikut untuk memetakan URL `/products` ke **ProductController**.

The image shows a VS Code editor window with the `Routes.php` file open. The file path is `D:\> tdd-codeigniter 5094-5076 > src > app > Config > Routes.php`. The code in the file is as follows:

```
1 <?php
2
3 use CodeIgniter\Router\RouteCollection;
4
5 /**
6 * @var RouteCollection $routes
7 */
8 $routes->get('/', 'Home::index');
9 $routes->resource('products', ['controller' => 'Product']);
```

4. **Jalankan Tes Kembali** Dengan semua komponen sudah di tempat, jalankan lagi perintah tes.

```
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Controllers/ProductControllerTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

. 1 / 1 (100%)

Time: 00:06.916, Memory: 18.00 MB

OK (1 test, 3 assertions)

Generating code coverage report in PHP format ... done [00:00.020]
Generating code coverage report in Clover XML format ... done [00:01.127]
Generating code coverage report in HTML format ... done [00:11.377]

Code Coverage Report:
 2025-07-28 03:37:55

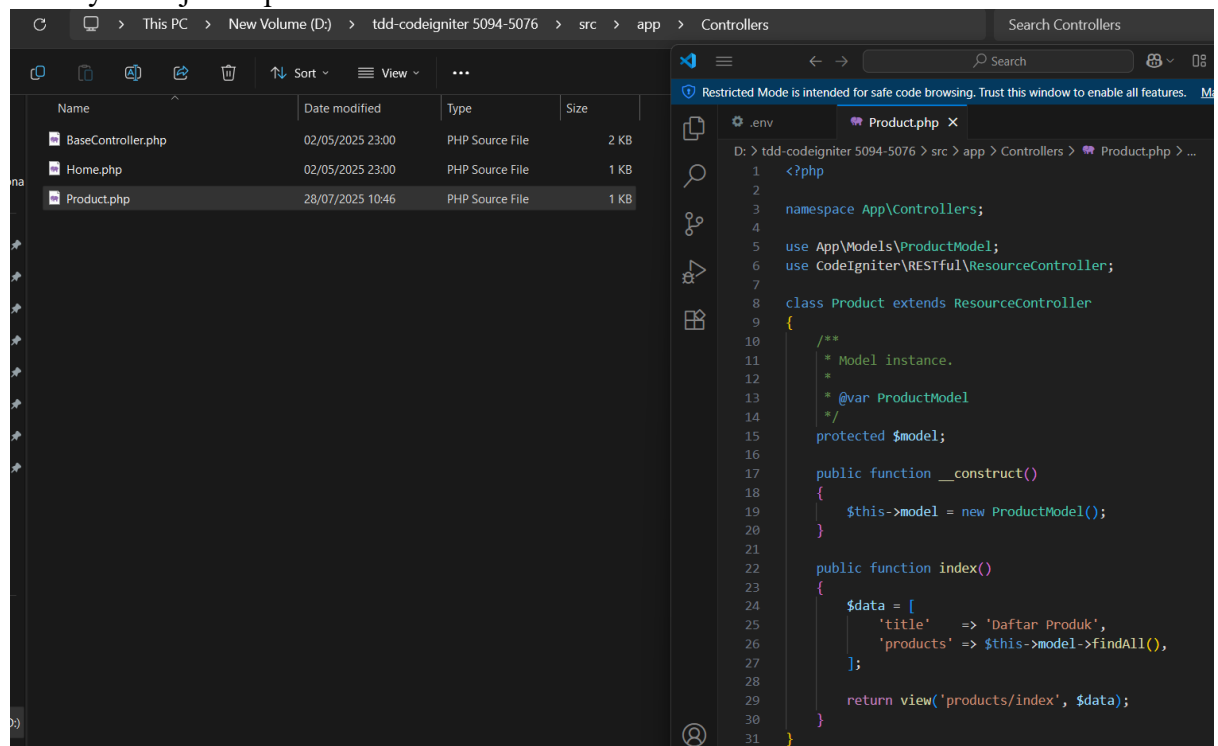
Summary:
Classes: 28.57% (2/7)
Methods: 22.22% (2/9)
Lines: 10.89% (11/101)

App\Controllers\Product
Methods: 100.00% (1/ 1) Lines: 100.00% (6/ 6)
Config\Database
Methods: 100.00% (1/ 1) Lines: 100.00% (3/ 3)
root@d0877f22f84c:/var/www/html#
```

Tes berhasil karena permintaan ke /products sekarang dapat diproses dengan benar oleh semua komponen yang telah kita buat. Tahap **GREEN** berhasil.

### Tahap 3: REFACTOR - Membersihkan dan Memperbaiki Kode

1. **Perbaiki Kode Controller** Buka kembali `src/app/Controllers/Product.php` dan ubah kodenya menjadi seperti ini:



```
D:\tdd-codeigniter 5094-5076 > src > app > Controllers > Product.php ...

1 <?php
2
3 namespace App\Controllers;
4
5 use App\Models\ProductModel;
6 use CodeIgniter\RESTful\ResourceController;
7
8 class Product extends ResourceController
9 {
10 /**
11 * Model instance.
12 *
13 * @var ProductModel
14 */
15 protected $model;
16
17 public function __construct()
18 {
19 $this->model = new ProductModel();
20 }
21
22 public function index()
23 {
24 $data = [
25 'title' => 'Daftar Produk',
26 'products' => $this->model->findAll(),
27];
28
29 return view('products/index', $data);
30 }
31 }
```

2. **Jalankan Seluruh Rangkaian Tes** Setelah melakukan refactoring, sangat penting untuk menjalankan **semua tes** untuk memastikan perubahan kita tidak merusak fungsionalitas yang ada, termasuk Unit Test yang kita buat sebelumnya.

```
root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

.....

Time: 00:07.305, Memory: 16.00 MB

OK (8 tests, 14 assertions)

Generating code coverage report in PHP format ... done [00:00.027]
Generating code coverage report in Clover XML format ... done [00:01.307]
Generating code coverage report in HTML format ... done [00:12.078]

Code Coverage Report:
 2025-07-28 04:06:50

Summary:
Classes: 28.57% (2/7)
Methods: 40.00% (4/10)
Lines: 11.88% (12/101)

App\Controllers\Product
Methods: 100.00% (2/ 2) Lines: 100.00% (6/ 6)
App\Database\Migrations\CreateProductsTable
Methods: 50.00% (1/ 2) Lines: 3.57% (1/ 28)
Config\Database
Methods: 100.00% (1/ 1) Lines: 100.00% (3/ 3)
```

Semua tes (baik Unit maupun Integration) harus tetap **HIJAU**. Ini membuktikan bahwa refactoring kita berhasil dan aman.

## E. TUGAS

Berdasarkan studi kasus yang telah diselesaikan, kembangkan fitur baru untuk melihat detail satu produk. Terapkan siklus TDD dalam pengerjaannya. Langkah-langkah Pengerjaan Tugas: Tugas ini juga tidak berubah. Ikuti langkah-langkah yang sama seperti pada modul sebelumnya untuk menyelesaikan fitur ini, dan pastikan semua perintah pengujian dijalankan dari dalam terminal kontainer.

1. **Tulis Tes Terlebih Dahulu (RED):** Buat metode tes `testShowFindsSingleProduct()` di `ProductControllerTest` yang mensimulasikan GET ke `/products/1` dan pastikan tes tersebut gagal.

```
<?php

namespace Tests\App\Controllers;

use CodeIgniter\Test\CIUnitTestCase;
use CodeIgniter\Test\ControllerTestTrait;
use CodeIgniter\Test\DatabaseTestTrait;
use App\Models\ProductModel;
use App\Controllers\Product;

class ProductControllerTest extends CIUnitTestCase
{
 use ControllerTestTrait, DatabaseTestTrait;

 protected $migrate = false;

 public function testShowFindsSingleProduct()
 {
 // ARRANGE
 $model = new ProductModel();
 $id = $model->insert([
 'name' => 'Produk Uji',
 'price' => 150.00
]);

 // ACT
 $result = $this->withURI("http://localhost:8080/products/{$id}")
 ->controller(Product::class)
 ->execute('show', $id);

 // ASSERT
 $this->assertTrue($result->isOk());
 $this->assertStringContainsString('Produk Uji', $result->getBody());
 }
}
```



```

root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Controllers/ProductControllerTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

E

Time: 00:09.503, Memory: 20.00 MB

There was 1 error:

1) Tests\App\Controllers\ProductControllerTest::testShowFindsSingleProduct
CodeIgniter\View\Exceptions\ViewException: Invalid file: "products/show.php"

/var/www/html/vendor/codeigniter4/framework/system/Exceptions/FrameworkException.php:39
/var/www/html/vendor/codeigniter4/framework/system/View/View.php:212
/var/www/html/vendor/codeigniter4/framework/system/Common.php:1173
/var/www/html/app/Controllers/Product.php:39
/var/www/html/vendor/codeigniter4/framework/system/Test/ControllerTestTrait.php:166
/var/www/html/tests/app/Controllers/ProductControllerTest.php:29

ERRORS!
Tests: 1, Assertions: 0, Errors: 1.

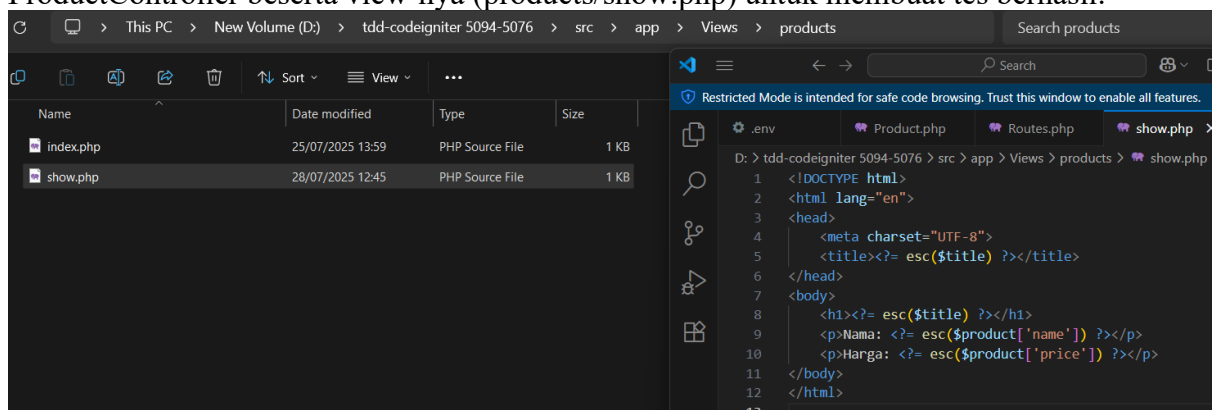
Generating code coverage report in PHP format ... done [00:00.032]
Generating code coverage report in Clover XML format ... done [00:01.459]
Generating code coverage report in HTML format ... done [00:12.647]

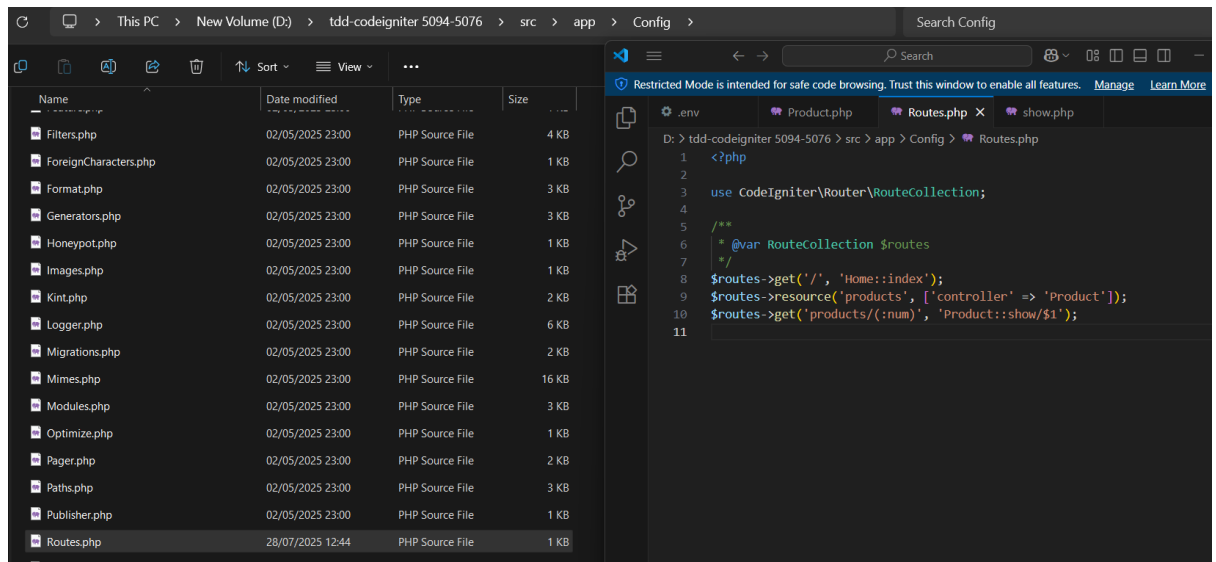
Code Coverage Report:
 2025-07-28 05:43:17

Summary:
Classes: 0.00% (0/7)
Methods: 0.00% (0/11)
Lines: 0.00% (0/108)

```

2. Implementasikan Fitur (GREEN): Buat metode `show($id = null)` di `ProductController` beserta view-nya (`products/show.php`) untuk membuat tes berhasil.





### 3. Jalankan Tes Lagi: Pastikan tes yang sebelumnya gagal kini berhasil.

```

root@d0877f22f84c:/var/www/html# ./vendor/bin/phpunit tests/app/Controllers/ProductControllerTest.php
PHPUnit 10.5.48 by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.29 with Xdebug 3.4.5
Configuration: /var/www/html/phpunit.xml.dist

.

Time: 00:06.452, Memory: 18.00 MB

OK (1 test, 2 assertions)

Generating code coverage report in PHP format ... done [00:00.023]
Generating code coverage report in Clover XML format ... done [00:00.836]
Generating code coverage report in HTML format ... done [00:12.289]

Code Coverage Report:
 2025-07-28 05:46:11

Summary:
Classes: 14.29% (1/7)
Methods: 18.18% (2/11)
Lines: 11.11% (12/108)

App\Controllers\Product
Methods: 33.33% (1/ 3) Lines: 53.85% (7/ 13)
Config\Database
Methods: 100.00% (1/ 1) Lines: 100.00% (3/ 3)

```

4. **Refactor (Jika Perlu):** Perbaiki kode tanpa mengubah fungsionalitasnya, dan pastikan semua tes tetap berhasil.

The screenshot displays a development environment with three main components:

- File Explorer:** Shows the directory structure of a CodeIgniter project. The files listed are `BaseController.php` (2 KB), `Home.php` (1 KB), and `Product.php` (1 KB).
- Code Editor:** Displays the `Product.php` file. The code defines a `Product` class extending `ResourceController`. It includes an `index` method that returns a view with product data and a `show` method that displays details for a specific product.
- Terminal:** Shows the execution of PHPUnit tests. The output indicates that 8 tests passed with 13 assertions. It also generates a code coverage report showing the following summary:

| Category                                    | Methods       | Lines         |
|---------------------------------------------|---------------|---------------|
| App\Controllers\Product                     | 0.00% (0/2)   | 50.00% (6/12) |
| App\Database\Migrations\CreateProductsTable | 50.00% (1/2)  | 3.57% (1/28)  |
| Config\Database                             | 100.00% (1/1) | 100.00% (3/3) |

## F. Kesimpulan :

Dalam praktikum ini berhasil diterapkan konsep **Test-Driven Development (TDD)** menggunakan CodeIgniter 4 pada lingkungan yang dikonfigurasi dengan Docker. Proses pengembangan dilakukan melalui tiga tahap utama yaitu **RED, GREEN, dan REFACTOR**. Tahap RED memastikan tes ditulis terlebih dahulu dan gagal sesuai harapan. Tahap GREEN menunjukkan bahwa implementasi kode minimal dapat membuat tes berhasil, sedangkan tahap REFACTOR memastikan kualitas kode tetap baik tanpa merusak fungsionalitas yang sudah ada. Selain itu, integrasi antara Unit Testing dan Integration Testing berhasil dilakukan untuk memastikan bahwa model, controller, dan view dapat bekerja bersama dengan benar. Melalui proses ini, diperoleh pemahaman lebih baik mengenai pentingnya pengujian otomatis untuk menjaga stabilitas aplikasi.